Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/05/2023
PI Name:	Baker, John Ph.D.		
Project Title:	Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53226-3548	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	02/04/2019	**End Date:**	02/03/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/03/2024 per NSSC information (Ed., 4/18/23) NOTE: End date changed to 02/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	December 2023 report: None.		
COI Name (Institution):	Kronenberg, Amy Sc.D. (Lawrence Berkeley National Laboratory) Willey, Jeffrey Ph.D. (Wake Forest University)		
Grant/Contract No.:	80NSSC19K0498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The interaction between simulated weightlessness and space radiation on cardiac disease needs to be understood prior to undertaking deep space exploratory missions. To date, combining the simulated spaceflight hazards of reduced weight bearing via hind limb unloading in rats of any strain with simultaneous GCR or gamma exposure in conscious animals has not been possible to conduct at the NASA Space Radiation Laboratory. To meet this challenge, we have worked with Jeffrey Willey, PhD, our collaborator at Wake Forest Medical School, and have engineered a solution that provides a realistic scenario for the investigation of exploratory spaceflight. Thus, our current ground-based studies directly address this important aspect of the NASA Human Research Program Degenerative Tissue Gaps as set forth in the Human Research Roadmap. The research currently underway is providing important information to help close these gaps. We determined the combined effects of simulated weightlessness and radiation exposure during deep space exploration on risk for and occurrence of cardiac disease in a ground-based rat model. These studies simulate a 3-3.5-year deep space human mission in male WAG/RijCmcr rats 6 months old at the start of the study, determining the impact of simulated weightlessness combined with components of GCRs using a simplified 5-ion mixed field beam with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for cardiac disease. Parallel studies were conducted with male rats from a second strain (Fischer 344) at an age representative of the astronaut corps. Specific Aim A: Simulate a 3-3.5 year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to photon radiation in male Fischer 344 rats with and without simulated weightlessness to estimate relative biological effectiveness (RBE) for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of X-rays: 3.0 Gy, 8.0 Gy, and 10.0 Gy alongside experimental hindlimb unloaded animals. Specific Aim C: Simulate a 3-3.5 year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth as it relates to patients who receive therapeutic radiation for the treatment of cancer and to individuals who receive accidental exposure to radiation from a nuclear accident.
Task Progress:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (Principal Investigator/PI Michael Delp, PhD; Radiation, "Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure"). This descoped application is designed to address key aspects of the research goals put forward in Topic 6 of NASA Research Announcement (NRA) NNJ16ZSA001N-SRHHC “Space Radiation and Human Health Countermeasures Topics”: 1) to provide detailed physiological and bio-molecular characterization of risk and prediction for cardiac disease from microgravity and space-like radiation doses that are mission relevant for future human spaceflight beyond low-Earth orbit and 2) to determine relative biological effectiveness (RBE) values for space radiation in appropriate animals using relevant intermediate as well as late physiological endpoints and effects. These experimentally derived RBE values will provide quantitative inputs into methods and models to calculate degenerative tissue responses in humans following exposures in space. The proposed studies center on the assessment of the degenerative tissue effects of space radiation on cardiac disease. The central hypothesis of this project is that simulated weightlessness interacts with whole-body exposure to space-relevant doses of radiation to potentiate risk for cardiac disease. Three specific aims are proposed to directly address the issue of risk for cardiac disease using experimental approaches in a rat model relevant to the research questions. The proposed investigations directly address important aspects of the NASA Human Research Program (HRP) Degenerative Tissue Gaps, as set forth in the Human Research Roadmap. HRP gaps are identified and addressed by each Specific Aim. The proposed research will provide important information to help close these gaps. Specific Aim A: Simulate a 3-3.5-year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to photon radiation in male Fischer 344 rats with and without simulated weightlessness to estimate RBE for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of x-rays: 3.0 Gy, 8.0 Gy, and 10.0 Gy alongside experimental hindlimb unloaded (HLU) animals. Specific Aim C: Simulate a 3-3.5-year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. In WAG/RijCmcr rats, our findings provide evidence for the occurrence of cardiac fibrosis resulting from exposure of rats to a 5-ion mixed field beam designed to broadly simulate the composition of the GCR behind shielding, with no additive effect of the simulated spaceflight hazard of reduced weight bearing via HLU on the impact of SimGCRSim. These findings were presented at the 2021 NASA Human Research Program Investigators' Workshop (IWS) in February 2021. In Fischer 344 rat's irradiation with SimGCRSim with and without hindlimb unloading had no effect on blood urea nitrogen or total cholesterol levels. Cardiac fibrosis was not detected either after GCRSim or with hindlimb unloading.

	These findings were presented at the 2022 NASA Human Research Program IWS in February 2022. In Fischer 344 rat's irradiation with photons with and without hindlimb unloading had no effect on blood urea nitrogen or total cholesterol levels. Cardiac fibrosis was not detected in Fischer 344 rats either after photon irradiation or with hindlimb unloading.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Lenarczyk M, Kronenberg A, Mäder M, Komorowski R, Hopewell JW, Baker JE. "Exposure to multiple ion beams, broadly representative of galactic cosmic rays, causes perivascular cardiac fibrosis in mature male rats." PLoS One. 2023 Apr 26;18(4):e0283877. https://doi.org/10.1371/journal.pone.0283877 ; PubMed PMID: 37099482; PubMed Central PMCID: PMC10132632 , Apr-2023
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